



CISCO PACKET TRACER

ULTIMATE CLI COMMAND CHEAT SHEET — HELLFIRE EDITION

ROUTERS & SWITCHES

CCNA LABS

Ghost Rider Command Reference

1. CISCO CLI MODES & NAVIGATION

Prompt	Mode	Command to Enter
Router>	User EXEC	Default on connect
Router#	Privileged EXEC	<code>enable</code>
Router(config)#	Global Config	<code>configure terminal</code> (<code>conf t</code>)
Router(config-if)#	Interface Config	<code>interface g0/0</code> (<code>int g0/0</code>)
Router(config-line)#	Line Config	<code>line console 0</code> / <code>line vty 0 4</code>

Navigation: `exit` (one step back) / `end` or `Ctrl+Z` (back to Privileged EXEC).

29. INTER-VLAN ROUTING (ROUTER-ON-A-STICK)

```
! Router Setup (Subinterfaces)
R1(config)# interface g0/0
R1(config-if)# no shutdown

R1(config)# interface g0/0.10
R1(config-subif)# encapsulation dot1q 10
R1(config-subif)# ip address 192.168.10.1 255.255.255.0

R1(config)# interface g0/0.20
R1(config-subif)# encapsulation dot1q 20
R1(config-subif)# ip address 192.168.20.1 255.255.255.0

! Connected Switch Port MUST be Trunk
S1(config)# interface g0/1
S1(config-if)# switchport mode trunk
```

2-6. BASIC DEVICE SECURITY & CONFIG

```
Router# conf t
Router(config)# hostname R1
R1(config)# enable secret cisco          !
Encrypted enable password (preferred)
R1(config)# enable password cisco        ! Plain-
text password (avoid)
R1(config)# service password-encryption ! Encrypt
all cleartext passwords

! Console Line Security
R1(config)# line console 0
R1(config-line)# password cisco
R1(config-line)# login
R1(config-line)# exit

! VTY Security (Telnet/SSH)
R1(config)# line vty 0 4
R1(config-line)# password cisco
R1(config-line)# login
R1(config-line)# exit

! Login Banner
R1(config)# banner motd # Unauthorized Access
Prohibited #
```

30-33. DHCP SERVER & RELAY

```
! Exclude Addresses (Configure First!)
R1(config)# ip dhcp excluded-address 192.168.10.1
192.168.10.10

! Create DHCP Pool
R1(config)# ip dhcp pool LAN_POOL
R1(dhcp-config)# network 192.168.10.0
255.255.255.0
R1(dhcp-config)# default-router 192.168.10.1
R1(dhcp-config)# dns-server 8.8.8.8

! DHCP Relay (When Server is on Remote Subnet)
R1(config)# interface g0/0
R1(config-if)# ip helper-address 192.168.100.10

! Verification
R1# show ip dhcp binding
R1# show ip dhcp pool
```

7–8. SAVE & ESSENTIAL VERIFICATION

```
! Saving Memory Structure: RAM (running-config) ->
NVRAM (startup-config)
R1# copy running-config startup-config    ! Short:
copy run start / wr / write memory

! Verification Commands
R1# show running-config                  ! Short:
show run
R1# show startup-config
R1# show ip interface brief            ! Short:
show ip int br (CRITICAL TOOL)
R1# show version                      ! Hardware/
IOS info
```

9–13. ROUTER INTERFACE CONFIGURATION

```
R1(config)# interface gigabitEthernet 0/0 ! Short:
int g0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# description LAN Connection
R1(config-if)# no shutdown              !
Enables interface (Short: no shut)

! Serial Interface Setup (DCE side requires clock
rate)
R1(config)# interface serial 0/0/0      ! Short:
int s0/0/0
R1(config-if)# ip address 10.0.0.1 255.255.255.252
R1(config-if)# clock rate 64000         ! Only
if DCE end
R1(config-if)# no shutdown
R1# show controllers serial 0/0/0       ! Verify
DCE/DTE status
```

- **up / up**: Interface working properly
- **admin down / down**: Shutdown state (forgot `no shut`)
- **down / down**: Physical layer or cable issue

14–16. STATIC & DEFAULT ROUTING

```
! Syntax: ip route DEST_NET SUBNET_MASK NEXT_HOP_IP
[or EXIT_INT]
R1(config)# ip route 192.168.2.0 255.255.255.0
10.0.0.2
R1(config)# ip route 192.168.2.0 255.255.255.0 g0/1

! Default Route (Gateway of Last Resort)
R1(config)# ip route 0.0.0.0 0.0.0.0 10.0.0.2

! Verify Routing Table
R1# show ip route
! Codes: C = Connected, L = Local, S = Static, O =
OSPF, D = EIGRP, R = RIP
```

34–35. NAT & PAT (PORT ADDRESS TRANSLATION)

```
! Static NAT
R1(config)# ip nat inside source static
192.168.1.10 200.1.1.10

! PAT / NAT Overload (Many-to-One)
R1(config)# access-list 1 permit 192.168.1.0
0.0.0.255
R1(config)# ip nat inside source list 1 interface
g0/1 overload

! Interface Assignments
R1(config)# interface g0/0
R1(config-if)# ip nat inside
R1(config)# interface g0/1
R1(config-if)# ip nat outside

! Verification
R1# show ip nat translations
R1# show ip nat statistics
```

36–37. ACCESS CONTROL LISTS (ACLs)

```
! Standard ACL (Filter near destination)
R1(config)# access-list 10 permit 192.168.1.0
0.0.0.255
R1(config)# interface g0/0
R1(config-if)# ip access-group 10 in

! Extended ACL (Filter near source)
R1(config)# access-list 100 deny tcp 192.168.1.0
0.0.0.255 host 192.168.2.10 eq 80
R1(config)# access-list 100 permit ip any any
R1(config)# interface g0/0
R1(config-if)# ip access-group 100 in
```

38–41. STP, PORTFAST & ETHERCHANNEL

```
! Spanning Tree (STP)
S1(config)# spanning-tree vlan 10 root primary
S1# show spanning-tree vlan 10

! PortFast & BPDU Guard (Access Ports)
S1(config)# interface fa0/1
S1(config-if)# spanning-tree portfast
S1(config-if)# spanning-tree bpduguard enable

! EtherChannel / LACP Integration
S1(config)# interface range g0/1 - 2
S1(config-if-range)# channel-group 1 mode
active ! LACP mode
S1(config)# interface port-channel 1
S1(config-if)# switchport mode trunk
S1# show etherchannel summary
```

17–18. RIP & OSPF DYNAMIC ROUTING

```
! RIP v2 Configuration
R1(config)# router rip
R1(config-router)# version 2
R1(config-router)# no auto-summary
R1(config-router)# network 192.168.1.0
R1(config-router)# network 10.0.0.0

! OSPF Configuration (Uses Wildcard Mask)
R1(config)# router ospf 1
R1(config-router)# router-id 1.1.1.1
R1(config-router)# network 192.168.1.0 0.0.0.255 area 0
R1(config-router)# network 10.0.0.0 0.0.0.3 area 0

! OSPF Verification
R1# show ip ospf neighbor
R1# show ip protocols
```

19–23. SWITCH CONFIG & MAC TABLE

```
! Switch Management IP (SVI)
S1(config)# interface vlan 1
S1(config-if)# ip address 192.168.1.2 255.255.255.0
S1(config-if)# no shutdown
S1(config)# ip default-gateway 192.168.1.1

! MAC Address Table
S1# show mac address-table
S1# clear mac address-table dynamic
```

24–28. VLANS & TRUNKING

```
! Create VLANs
S1(config)# vlan 10
S1(config-vlan)# name SALES
S1(config)# vlan 20
S1(config-vlan)# name IT

! Access Port Assignment
S1(config)# interface fa0/1
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 10

! Range Assignment
S1(config)# interface range fa0/1 - 10
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 10

! Trunk Port Setup
S1(config)# interface g0/1
S1(config-if)# switchport mode trunk
S1(config-if)# switchport trunk native vlan 99
S1(config-if)# switchport trunk allowed vlan 10,20,30
S1(config-if)# switchport trunk allowed vlan add 40
S1(config-if)# switchport trunk allowed vlan remove 20

! Verification
S1# show vlan brief
S1# show interfaces trunk
```

42–45. HSRP, VRRP, CDP & LLDP

```
! HSRP Configuration (Gateway Redundancy)
R1(config-if)# standby 1 ip 192.168.1.254
R1(config-if)# standby 1 priority 110
R1(config-if)# standby 1 preempt
R1# show standby

! CDP & LLDP Neighbor Discovery
R1(config)# cdp run      | S1(config)# lldp run
R1# show cdp neighbors | S1# show lldp neighbors detail
```

48–52. SSH, DNS & IPV6 BASICS

```
! Secure Shell (SSH) Config
R1(config)# hostname R1
R1(config)# ip domain-name example.com
R1(config)# crypto key generate rsa ! Select 1024+ bits
R1(config)# username admin secret cisco
R1(config)# line vty 0 4
R1(config-line)# login local
R1(config-line)# transport input ssh
R1(config)# ip ssh version 2

! Disable Annoying Domain Lookup Hangs
R1(config)# no ip domain-lookup

! IPv6 Routing & Interface Setup
R1(config)# ipv6 unicast-routing
R1(config)# interface g0/0
R1(config-if)# ipv6 address 2001:db8:1::1/64
R1(config-if)# ipv6 address fe80::1 link-local
R1(config)# ipv6 route 2001:db8:2::/64 2001:db8:1::2
R1# show ipv6 interface brief | show ipv6 route
```

53–54. PC COMMAND PROMPT COMMANDS

Task	PC CLI Command
Check IP Setup	<code>ipconfig</code> / <code>ipconfig /all</code>
Renew DHCP	<code>ipconfig /release</code> then <code>/renew</code>
Test Link / Route	<code>ping 192.168.1.1</code> <code>tracert 8.8.8.8</code>
DNS Query	<code>nslookup google.com</code>

57–58. CLEAN SLATE: REMOVAL & FACTORY RESET

```
! Undo Command Rule: Add 'no' before original command
R1(config)# no ip address | no vlan 10 | no ip route ...

! Factory Reset Switch / Router
Switch# erase startup-config ! Or: write erase
Switch# delete vlan.dat      ! Clear VLAN database on switches
Switch# reload                ! Reboot device
```

🔥 PACKET TRACER MASTER REFERENCE & TROUBLESHOOTING

GHOST RIDER EDITION

55. ESSENTIAL VERIFICATION MATRIX

Area	Command
Router IP	<code>show ip interface brief</code>
Routing Table	<code>show ip route</code>
Switch VLANs	<code>show vlan brief</code>
Switch Trunks	<code>show interfaces trunk</code>
MAC Table	<code>show mac address-table</code>
STP Info	<code>show spanning-tree</code>
EtherChannel	<code>show etherchannel summary</code>
Neighbors	<code>show cdp neighbors</code>
DHCP Leases	<code>show ip dhcp binding</code>
NAT Table	<code>show ip nat translations</code>
OSPF Peers	<code>show ip ospf neighbor</code>
HSRP State	<code>show standby</code>

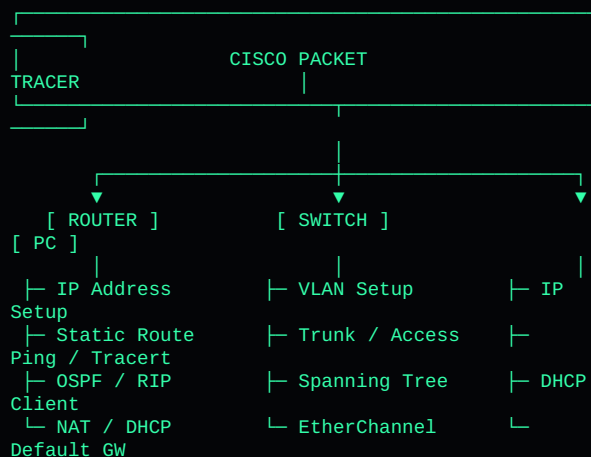
59. ESSENTIAL CLI SHORTCUTS

Key Combo	Action / Function
<code>Tab</code>	Auto-completes partial command
<code>?</code>	Displays available options
<code>Ctrl + C</code>	Cancel current command/exit mode
<code>Ctrl + Z</code>	Jump straight to Privileged EXEC
<code>Ctrl + Shift + 6</code>	Interrupt DNS lookup freeze
<code>Up / Down Arrow</code>	Scroll through command history
<code>Ctrl + A / Ctrl + E</code>	Move cursor to Line Start / End
<code>Ctrl + U / Ctrl + W</code>	Erase Line / Erase Word

61. HELLFIRE TROUBLESHOOTING FLOWCHART

1. PC IP / Subnet / GW (`ipconfig /all`)
- ↓
2. Ping Local Gateway (`ping 192.168.1.1`)
- ↓
3. Check Switch VLAN & Access Port (`sh vlan br`)
- ↓
4. Check Switch Trunk Links (`sh int trunk`)
- ↓
5. Router Interface Status (`sh ip int br`)
- ↓
6. Router Routing Table (`sh ip route`)
- ↓
7. Verify ACLs & NAT Setup (`sh ip nat trans`)

63. VISUAL FUNCTIONAL ARCHITECTURE



⚡ THE GOLDEN RULE OF PACKET TRACER LABS ⚡

1. CONFIGURE FIRST
2. VERIFY WITH SHOW COMMANDS
3. TEST WITH PING & TRACEROUTE

